Literature targets for the one-loop inner product and $Q^\dagger = S$

Step 2, directive R5

2026-10-28

1 Labels, names, sources

Labels. [S] transcribed from the source at the cited equation (arXiv PDF of the stated version, read in full at the cited places); [D] derived here from quoted formulas, proof given, checked by `step2/literature/check_ip_targets.py` (output `check_ip_targets_output.txt`, items C1–C8); [I] identification or inference stated in no source.

Companion. Target n of this file is entry `IP`n of `ip_targets.json` (built by `build_ip_targets_json.py`); Proposition 1 is `IP-P1`. Code names of the Gram matrices occur only there.

Name clash. $Q_1, S_1, D_1, K_1, P_1, M_1, R_1$ are the one-loop components of $(d_{CE} + Q)O[\infty]$ along $\xi \in \Sigma$, $\eta \in \Sigma^*$, $\Delta = 2E$, the special conformal field, translations, rotations of V, $\mathfrak{so}(W^\perp_{\mathbb{R}})$ (`step2/theory/odd_components_design.tex`, def:o-comp). The manuscript's spinor parameters $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ do not occur: elements of Σ, Σ^* are written $\xi = (\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$, $\eta = (\eta^\alpha{}_i, \bar\eta^{\dot\alpha i})$ (dictionary, entry `parameters`). Spin-chain generators of [B03], [B04], [Z05], [Zw11] are coefficients of powers of their couplings and carry the source key, e.g. $Q_2^{[\mathrm{B03}]}$. Q without subscript is the linear BV differential. The radial conjugation of the user's directive is written $X \mapsto X^\dagger$; the complex-conjugate transpose of a matrix is written $\overline{X}^{\mathrm{T}}$. The inner product of local operators a, b is the bra-ket of [SD] (113)–(114), $\langle a|b\rangle = \lim_{|y|\to\infty} |y|^{2\times(\text{dimension of } a)} \langle 0|a(y)^\dagger b(0)|0\rangle$; $\langle a|b\rangle\big|_{\text{tree}}$ and $\langle a|b\rangle\big|_{\text{one loop}}$ are its parts of order $\hbar^0$ and $\hbar^1$ relative to the tree value. The odd pairing $\langle e, Qe\rangle$ of the manuscript is never meant by a bra-ket.

Two-component conventions. Dictionary `step2/notation/dictionary.json`: $\varepsilon^{12} = +1$; Euclidean $\sigma^m = (\vec\sigma, i)$, $\bar\sigma^m = (-\vec\sigma, i)$, $\sigma^m\bar\sigma^n + \sigma^n\bar\sigma^m = -2\delta^{mn}$, $\bar\sigma^m = -\overline{\sigma^m}^{\mathrm{T}}$; fields A_m, $A_{ij} = A_I\Sigma^I_{ij}$, $A^{ij} = \frac{1}{2}\varepsilon^{ijkl}A_{kl}$ (real slice: complex conjugate of A_{ij} is A^{ij}), $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$; pairing $(\xi, \eta) = \xi_\alpha{}^i\eta^\alpha{}_i - \bar\xi_{\dot\alpha i}\bar\eta^{\dot\alpha i}$; Killing spinor $\epsilon_\alpha{}^i = \xi_\alpha{}^i - ix_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i} = \bar\xi_{\dot\alpha i} + ix_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$; generator labels $\xi_\alpha{}^i \leftrightarrow Q^i{}_\alpha$ (block $(3,1)$ of [SW] (17)), $\bar\xi_{\dot\alpha i} \leftrightarrow \bar Q_{\dot\alpha i}$ $(2,3)$, $\eta^\alpha{}_i \leftrightarrow S^\alpha{}_i$ $(1,3)$, $\bar\eta^{\dot\alpha i} \leftrightarrow \bar S^{\dot\alpha i}$ $(3,2)$; translation $(2,1)$, special conformal $(1,2)$; operators: Konishi $= \frac{1}{4}\int\omega(A^{ij}, A_{ij})$, $\mathbf{20'} = \frac{1}{16}\int\omega(w_1^{ij}A_{ij}, w_2^{kl}A_{kl}) - \frac{1}{96}w_1^{ij}w_{2ij}\int\omega(A^{kl}, A_{kl})$, $\iota_w A = \frac{1}{4}w^{ij}A_{ij}$; $\mathfrak{su}(N)$: $A_I = -igA_I^aT^a$, $\lambda = -\frac{ig}{\sqrt{2}}\lambda^aT^a$, $(X,Y) = -\frac{4\hbar}{g^2}\operatorname{tr}(XY) = 2\hbar X^aY^a$, $\dfrac{\hbar\,\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)} = -\dfrac{g^2N}{32\pi^2}$ (C7). Dilatation on operators (ch06 cor:c6-dilcomp): component along $\Delta = 2E$ equals $2(E + w_O + \text{coefficient})$, scaling dimension $w_O + \text{coefficient}$.

Existing entries (not duplicated, cited by id). `n4_untwisted.json`: sources B03, B03c, B04, Z05, CY13, CL22, GGKM08, EJSS05, BKRS01; derived D2 (similarity moves [B03] one-loop supercharge corrections between Q and S), D4 ([Z05] frame with uncorrected translations), D5 ([B03] (4.13) erratum and eigen-operators), D6, D7. `q1_targets.json`: T5 (Konishi entry), T7 (no Leibniz rule), T11 (scheme statements). `sweep.json`: paper ids 2306.01039, 2512.07771, 2306.04693, hep-th/0310252, hep-th/0307015, hep-th/0511109, 1111.0083 (abstract only there), 1305.6314, 0803.4183, 2209.06728, hep-th/0104016, hep-th/0501077.

Source keys (version read; local copies in `step2/literature/ip_sources/`).

[Mi97] S. Minwalla, *Restrictions imposed by superconformal invariance on quantum field theories*, hep-th/9712074v2.

[DO02] F. A. Dolan, H. Osborn, *On short and semi-short representations for four-dimensional superconformal symmetry*, hep-th/0209056v2.

[KMMR] J. Kinney, J. Maldacena, S. Minwalla, S. Raju, *An index for 4 dimensional super conformal theories*, hep-th/0510251v4.

[SD] D. Simmons-Duffin, *TASI lectures on the conformal bootstrap*, 1602.07982v1.

[Ry] S. Rychkov, *EPFL lectures on conformal field theory in $D \geq 3$ dimensions*, 1601.05000v2.

[B03] N. Beisert, *The su(2|3) dynamic spin chain*, hep-th/0310252v2.

[B04] N. Beisert, dissertation, hep-th/0407277v4.

[BKS] N. Beisert, C. Kristjansen, M. Staudacher, *The dilatation operator of $\mathcal{N} = 4$ SYM*, hep-th/0303060v4.

[Zw11] B. I. Zwiebel, *From scattering amplitudes to the dilatation generator in $\mathcal{N} = 4$ SYM*, 1111.0083v2.

[DFS] E. D'Hoker, D. Z. Freedman, W. Skiba, *Field theory tests for correlators in the AdS/CFT correspondence*, hep-th/9807098v1.

[PSZ] S. Penati, A. Santambrogio, D. Zanon, *Two-point functions of chiral operators in $\mathcal{N} = 4$ SYM at order g^4*, hep-th/9910197v1.

[EHW] B. Eden, P. S. Howe, P. C. West, *Nilpotent invariants in $\mathcal{N} = 4$ SYM*, hep-th/9905085v1.

[HSW] P. S. Howe, E. Sokatchev, P. C. West, *3-point functions in $\mathcal{N} = 4$ Yang–Mills*, hep-th/9808162v3 (abstract and introduction).

[AFGJ] D. Anselmi, D. Z. Freedman, M. T. Grisaru, A. A. Johansen, *Nonperturbative formulas for central functions of supersymmetric gauge theories*, hep-th/9708042v1.

[BRvR] C. Beem, L. Rastelli, B. C. van Rees, *The $\mathcal{N} = 4$ superconformal bootstrap*, 1304.1803v2.

[BKRS] M. Bianchi, S. Kovacs, G. Rossi, Y. S. Stanev, *Properties of the Konishi multiplet in $\mathcal{N} = 4$ SYM theory*, hep-th/0104016v1.

[EJSS] B. Eden, C. Jarczak, E. Sokatchev, Ya. S. Stanev, *Operator mixing in $\mathcal{N} = 4$ SYM: the Konishi anomaly revisited*, hep-th/0501077v2.

[BERS] M. Bianchi, B. Eden, G. Rossi, Y. S. Stanev, *On operator mixing in $\mathcal{N} = 4$ SYM*, hep-th/0205321v1.

[Ryz] A. V. Ryzhov, *Quarter BPS operators in $\mathcal{N} = 4$ SYM*, hep-th/0109064v2.

[DHHR] E. D'Hoker, P. Heslop, P. Howe, A. V. Ryzhov, *Systematics of quarter BPS operators in $\mathcal{N} = 4$ SYM*, hep-th/0301104v2.

[BKPSS] N. Beisert, C. Kristjansen, J. Plefka, G. W. Semenoff, M. Staudacher, *BMN correlators and operator mixing in $\mathcal{N} = 4$ SYM*, hep-th/0208178v2.

[BMV] K. Budzik, H. Murali, P. Vieira, *Following black hole states*, 2306.04693v2.

[GGKM] L. Grant, P. A. Grassi, S. Kim, S. Minwalla, *Comments on 1/16 BPS quantum states and classical configurations*, 0803.4183v1.

[CY] C.-M. Chang, X. Yin, *1/16 BPS states in $\mathcal{N} = 4$ SYM*, 1305.6314v2.

[CL22] C.-M. Chang, Y.-H. Lin, *Words to describe a black hole*, 2209.06728v4.

[BK] K. Budzik, J. Kulp, *Loop corrected supercharges from holomorphic anomalies*, 2512.07771v3.

[BGKWWY] K. Budzik, D. Gaiotto, J. Kulp, B. R. Williams, J. Wu, M. Yu, *Semi-chiral operators in 4d $\mathcal{N} = 1$ gauge theories*, 2306.01039v2.

[KKLP] S. Kim, S. Kim, S. Lee, J. Park, *Quantum black hole cohomologies*, 2606.27955v2.

[KLLS] S. Kim, J. Lee, S. Lee, J. Song, *Black hole cohomologies in $\mathcal{N}=4$ SYM*, 2610.20340v1.

[Z05] B. I. Zwiebel, hep-th/0511109v3 (through `n4_untwisted.json`).

Regulators of DR/DRED/SSDR papers are recorded; their statements are statements only.

2 The conjugation of radial quantisation

Target 1 (Euclidean generators, radial hermiticity)**.** [S] [Mi97] (2.20)–(2.27), (2.32)–(2.33):

$$M'_{pq}=S_{pq},\quad D'=iS_{-1\,0},\quad P'_p=S_{p\,-1}+iS_{p0},\quad K'_p=S_{p\,-1}-iS_{p0};$$
$$M'^{\dagger}=M',\quad D'^{\dagger}=-D',\quad P'^{\dagger}=K',\quad K'^{\dagger}=P',$$

scaling dimension ϵ_0 from $[D',O']=(-i)(-i\epsilon_0 O')$; unitarity at level one $\Leftrightarrow \langle\{s\}|K'_\mu P'_\nu|\{t\}\rangle = 2\langle\{s\}|\epsilon_0+(-i)M'_{\mu\nu}|\{t\}\rangle\geq 0$.

Regulator. none (representation theory).

Scheme. independent.

Project object. restriction of $X\mapsto X^\dagger$ to the conformal Killing fields (Target 6); positivity of $\langle a|b\rangle\big|_{\text{tree}}$ on level-one descendants and its one-loop correction (Target 22).

Two-component. $P_{\alpha\dot\alpha}=\sigma^m_{\alpha\dot\alpha}P_m$, $K^{\dot\alpha\alpha}=\bar\sigma^{m\dot\alpha\alpha}K_m$ for the generators along e_m; Target 6 gives $(P_{\alpha\dot\alpha})^\dagger = K^{\dot\alpha\alpha}$.

Target 2 (Euclidean supercharges)**.** [S] [Mi97] (3.2)–(3.3), (3.28)–(3.29): $Q'=\frac{1}{\sqrt2}(Q-i\sigma_0 S)$, $S'=\frac{1}{\sqrt2}(Q+i\sigma_0 S)$, Majorana basis with $Q^\dagger=Q$, $S^\dagger=S$ (Lorentzian); radially $Q'^\dagger=S'$, $S'^\dagger=Q'$. §5.2: Tr of p scalars, $\epsilon_0=p$, $SU(4)$ labels $(0,p,0)$, is the special short representation at the threshold $\epsilon_0=p$.

Regulator. none.

Scheme. independent.

Project object. odd part of $X\mapsto X^\dagger$: blocks $(3,1)\leftrightarrow(1,3)$, $(2,3)\leftrightarrow(3,2)$ (Target 7).

Two-component. $(Q^i{}_\alpha)^\dagger = S^\alpha{}_i$, $(\bar Q_{\dot\alpha i})^\dagger=\bar S^{\dot\alpha i}$; on parameters Target 7.

Target 3 (Bras by inversion; $2\{S,Q\}$)**.** [S] [KMMR] text before (2.2), (2.2), (2.9)–(2.10), (3.1)–(3.2), §5: bras are kets acted on by the inversion $x^\mu\mapsto x^\mu/x^2$, under which $M\mapsto M$, $H\mapsto -H$, $P\mapsto K$;

$$M^\dagger_{\mu\nu}=M_{\mu\nu},\quad P_\mu=K^\dagger_\mu,\quad (Q)^\dagger=S,\quad (\bar Q)^\dagger=\bar S,$$
$$2\{S,Q\}=H-2J_1-2\sum_{k=1}^{m-1}\tfrac{m-k}{m}R_k-\tfrac{(4-m)r}{2m}\equiv\Delta^{[\text{KMMR}]}\geq 0,$$

$|\psi\rangle = Q\frac{S}{\Delta_0}|\psi\rangle + S\frac{Q}{\Delta_0}|\psi\rangle$ at $\Delta^{[\text{KMMR}]}=\Delta_0>0$; §5: $Q\sim d$, $S\sim d^*$, $\Delta^{[\text{KMMR}]}\sim dd^*+d^*d$, harmonic states $\leftrightarrow$ Q-cohomology.

Regulator. none.

Scheme. independent.

Project object. for every odd x_1, the dilatation part of $[x_1,x_1^\dagger]$ has one sign on all four odd blocks (Target 7); the Hodge decomposition with $\langle\,|\,\rangle\big|_{\text{tree}}+\hbar\langle\,|\,\rangle\big|_{\text{one loop}}$ (Target 27).

Two-component. $m=4$, $Q=Q^4{}_-$: $2\{Q,Q^\dagger\}=E-2J-H_1-H_2-H_3$ in the Cartan basis of [CY] (2.1); labels as in Target 2.

Target 4 (Lorentzian hermiticity and the conformal map to the radial frame)**.** [S] [DO02] (2.11)–(2.17), (3.12)–(3.16), (A.8)–(A.11): $M^\dagger_{AB}=M_{AB}$, $(Q^i{}_\alpha)^\dagger=\bar Q_{i\dot\alpha}$, $(S_i{}^\alpha)^\dagger=\bar S^{i\dot\alpha}$; the similarity $e^{-\frac{\pi}{2}M_{0d}}(\cdot)e^{\frac{\pi}{2}M_{0d}}$ maps $iD\mapsto -H$, $P\mapsto E^+$, $K\mapsto E^-$ with $E^{-r}=(E^{+r})^\dagger$;

$$Q^{+i}{}_\alpha=\tfrac{1}{\sqrt2}(Q^i{}_\alpha+\sigma_{0\alpha\dot\alpha}\bar S^{i\dot\alpha}),\quad Q^{-\alpha}{}_i=\tfrac{1}{\sqrt2}(S_i{}^\alpha+\bar Q_{i\dot\alpha}\bar\sigma_0^{\dot\alpha\alpha}),\qquad (Q^{+i}{}_\alpha)^\dagger=Q^{-\beta}{}_i\sigma_{0\beta\dot\alpha},$$

$\{Q^{+i}{}_\alpha, Q^{-\beta}{}_j\} = 2\delta^i_j\delta^\beta_\alpha H + 4\delta^i_j \tilde{M}_\alpha{}^\beta - 4\delta^\beta_\alpha R^i{}_j$; primary norm ${}_{\bar{I}}\langle\Delta|\Delta\rangle_I = N_{\bar{I}I}$ positive definite; two-point function $\langle 0|\bar{O}_{\bar{I}}(0)O_I(x)|0\rangle = I_{\bar{I}I}(x)/(x^2)^\Delta$, $I(x) = N\exp(-\tan^{-1}\frac{r}{\tau}\,2s_{0\hat{a}}n_{\hat{a}})$; spinor: $I(x) = -i\sigma^a x_a/(x^2)^{1/2}$ with $N = \sigma_0$; vector: $I_{cd} = \eta_{cd} - 2x_c x_d/x^2$.

Regulator. none.

Scheme. independent.

Project object. the conformal map generated by P and K ($M_{0d} = \frac{1}{2}(P+K)$-direction) relating a real structure with a reflection to the radial one (R5(a)); the spinor inversion matrix in the field realisation of $X \mapsto X^\dagger$ on λ (Target 5).

Two-component. Euclidean: spinor inversion matrix $x_m\sigma^m_{\alpha\dot\alpha}/|x|$, inverse $-x_m\bar\sigma^{m\dot\alpha\alpha}/|x|$ since $(x\cdot\sigma)(x\cdot\bar\sigma) = -x^2$.

Target 5 (Field conjugation; reflection and inversion)**.** [S] [SD] (97), (99), (106), (108)–(117); [Ry] (3.34)–(3.45):

$$O_E(t_E,x)^\dagger = \Theta\cdot O_E(-t_E,x), \qquad O(x)^\dagger = x^{-2\Delta}\,I(x)\cdots I(x)\,O\big(\tfrac{x}{x^2}\big), \quad I^\mu{}_\nu = \delta^\mu_\nu - \tfrac{2x^\mu x_\nu}{x^2},$$

(Δ the scaling dimension and I the inversion tensor of [SD]); charges $Q^\dagger = -Q^{II}$; $M^\dagger_{\mu\nu} = -M_{\mu\nu}$, $D^\dagger = D$, $P^\dagger_\mu = K_\mu$ ([SD] generators without i); $\langle O|K_\mu P_\nu|O\rangle = 2\Delta\delta_{\mu\nu}\langle O|O\rangle$, $\langle O(y)O(x)\rangle = y^{-2\Delta}\langle O|O\rangle(1 + 2\Delta\frac{y\cdot x}{y^2} + \dots)$. [Ry]: N–S quantisation with K_1+P_1, conjugate to D by a rotation by $\pi/2$ in $SO(D+1,1)$; the special conformal map (3.35) and a translation carry the N–S foliation to the radial one; the reflection $\Theta : x_1 \mapsto -x_1$ gives $[\phi(x)]^\dagger = \phi(\Theta x)$ (3.37) and is carried to the inversion R (3.40); $[\phi(x)]^\dagger = r^{-2\Delta_\phi}\phi(Rx)$ (3.41); $K_\mu = RP_\mu R$, $P_\mu = K^\dagger_\mu$ (3.43)–(3.45).

Regulator. none.

Scheme. independent.

Project object. R5(a) [I]: the reflection in a hyperplane is a linear isometry of $\mathbb{R}^4$, hence commutes with $H = d^*d\otimes\mathrm{id}$, K_t, $P(\varepsilon,\infty)$; it reverses the orientation, so it exchanges S_+ and S_- (remark after prop:c1-chirality), and composed with the antilinear real-slice conjugation ($W \leftrightarrow W^*$) it preserves the chirality content $S_+\otimes W \oplus S_-\otimes W^*$ of Σ, the action and the rotation-covariant $I^{CT}[\varepsilon]$; the radial conjugation is its conjugate by the conformal map of [Ry] (3.35), so its one-loop realisation is fixed by the reflection together with P_1, K_1, D_1.

Two-component. [I] weights from ex:c2-dil: $A_{ij}(x) \mapsto |x|^{-2}A^{ij}(x/x^2)$; $A_m(x) \mapsto |x|^{-2}(\delta_m{}^n - 2x_m x^n/x^2)\overline{A_n}(x/x^2)$; $\lambda_\alpha{}^i(x) \mapsto |x|^{-4}x_n\sigma^n_{\alpha\dot\alpha} \times$ (conjugate dotted field)$^{\dot\alpha i}(x/x^2)$ up to a phase fixed by the real slice (R5(a)).

Target 6 (Inversion on the conformal Killing fields of the realisation)**.** [D] With $R(x) = x/(x,x)$ and R_* the pushforward of vector fields (C6):

$$R_*\big(x\mapsto b\big) = \big(x\mapsto (x,x)b - 2(b,x)x\big), \qquad R_*\big((x,x)b - 2(b,x)x\big) = b,$$
$$R_*(E) = -E, \qquad R_*(\text{rotation}) = \text{rotation}.$$

The antilinear map $v \mapsto -R_*(\bar v)$ reverses the bracket $[v,v'] = v(v') - v'(v)$ of def:c2-even, fixes $\Delta = 2E$, maps the translation along b to the special conformal field along $-\bar b$, and maps a rotation X_{mn} to $-\overline{X_{mn}}$. With [prop:c2-special] $[x\mapsto v,\ (x,x)b - 2(b,x)x] = -(b,v)\Delta + (\text{rotation})$, the dilatation part of $[P,P^\dagger]$ is $+|b|^2\Delta$ for the translation along b.

Proof. $dR_x(e) = e/x^2 - 2x(x,e)/x^4$ at $x = y/(y,y)$ is $(y,y)e - 2(e,y)y$; the remaining identities and the bracket reversal on the pairs $(P,K),(E,P),(E,K),(\mathrm{rot},K),(P,\mathrm{rot})$ are exact sympy evaluations (C6); R is an involution, and $-R_*(\overline{[v,w]}) = [-R_*\bar w, -R_*\bar v]$ because R_* preserves brackets.

Regulator. none.

Scheme. independent.

Project object. the even part of $X \mapsto X^\dagger$ on $\mathfrak{psl}(4|4)$ of Chapter 2; consistent with [SD] (112), [Ry] (3.43)–(3.45), [Mi97] (2.24)–(2.27).

Two-component. bispinor $b_{\alpha\dot\alpha} = \sigma^m_{\alpha\dot\alpha}b_m$; the image parameter $-\bar b_m$ has $\bar\sigma^{m\dot\alpha\alpha}(-\bar b_m) = \overline{b_{\alpha\dot\alpha}}$ (from $\overline{\sigma^m_{\alpha\dot\alpha}} = -\bar\sigma^{m\dot\alpha\alpha}$): $(P_{\alpha\dot\alpha})^\dagger = K^{\dot\alpha\alpha}$.

Target 7 (The conjugation on $\mathfrak{psl}(4|4)$)**.** [D] In the block presentation of [SW] (17) with a basis unitary for the compact $\mathrm{Spin}(4) \times SU(4)$ Hermitian forms of S_+, S_-, W:

$$X^\dagger = \mathrm{diag}(-1_2, 1_2 \,|\, 1_4)\, \overline{X}^{\mathrm{T}}\, \mathrm{diag}(-1_2, 1_2 \,|\, 1_4).$$

(a) $X \mapsto X^\dagger$ is antilinear and $[X, Y\}^\dagger = [Y^\dagger, X^\dagger\}$ for the supercommutator; (b) blocks $(2,1) \leftrightarrow (1,2)$, $(3,1) \leftrightarrow (1,3)$, $(2,3) \leftrightarrow (3,2)$, diagonal blocks to themselves, $\Delta^\dagger = \Delta$; (c) for unit elements, the coefficient of Δ in $[X, X^\dagger\}$ (supertrace against Δ over the supertrace of Δ^2) is $-\frac{1}{4}$ on all four odd blocks, $+\frac{1}{2}$ on $(2,1)$, $-\frac{1}{2}$ on $(1,2)$; (d) it is the transport by the supertranspose of the oscillator conjugation of [B04] (D.16)–(D.19) ($\mathcal{A} = (\mathbf{a}, \mathbf{b}^\dagger | \mathbf{c})$, $\mathcal{A}^\dagger = (\mathbf{a}^\dagger, -\mathbf{b} | \mathbf{c}^\dagger)$, $(J_A{}^B)^\dagger = \pm J_B{}^A$ with sign $-$ exactly when one index is in the $\mathbf{b}$ block), the supertranspose carrying $D^{[\mathrm{B04}]} = 1 + \frac{1}{2}\mathbf{a}^\dagger\mathbf{a} + \frac{1}{2}\mathbf{b}^\dagger\mathbf{b}$ to $\Delta/2$ and reversing brackets up to the Koszul sign; (e) $\mathrm{diag}(1_2, -1_2 | 1_4)$ and the identity give the patterns $(+\frac{1}{2}, -\frac{1}{2}; +\frac{1}{4}, \dots)$ and $(-\frac{1}{2}, +\frac{1}{2}; \pm\frac{1}{4})$, excluded by a $*$-representation, in which $[P, P^\dagger]$ and $\{x_1, x_1^\dagger\}$ have dilatation parts of opposite sign and all $\{x_1, x_1^\dagger\}$ the same sign ([KMMR] (3.1), [DO02] (3.14), [SD] (112)); a block-diagonal Hermitian matrix with the same signature pattern gives the same conclusions (C5).

Proof. (a)–(e) are exact matrix identities on $\mathfrak{gl}(4|4)$ evaluated on random homogeneous supermatrices and unit elements (C5); (d): $\big((\mathcal{A}^B)^\dagger\big) = \pm\mathcal{A}^\dagger_B$ with $-$ on $\mathbf{b}$, so $(J_A{}^B)^\dagger = (\mathcal{A}^\dagger_A \mathcal{A}^B)^\dagger = \pm \pm \mathcal{A}^\dagger_B \mathcal{A}^A$.

Regulator. none.

Scheme. independent.

Project object. the antilinear anti-automorphism of R5(a) in the realisation of Chapter 2; with Target 6 on the even part.

Two-component. [D] (C8) On parameters, $\xi \mapsto \xi^\dagger \in \Sigma^*$ with

$$\eta^\alpha{}_i = -\overline{\xi_\alpha{}^i}, \qquad \bar\eta^{\dot\alpha i} = \overline{\bar\xi_{\dot\alpha i}},$$

up to the outer automorphism multiplying $S_+ \otimes W$, $S_- \otimes W$ by a positive number and $S_- \otimes W^*$, $S_+ \otimes W^*$ by its inverse: $[\xi_1, \xi_2]^\dagger = [\xi_2^\dagger, \xi_1^\dagger]$ with $[\xi_1, \xi_2] = \Gamma(\xi_1, \xi_2)|_V$, $[\eta_1, \eta_2] = -(\text{special conformal field along } \Gamma(\eta_1, \eta_2)|_V)$ [prop:c2-special] and Target 6 requires $\Gamma(\xi_2^\dagger, \xi_1^\dagger)|_V = \overline{\Gamma(\xi_1, \xi_2)|_V}$, which holds iff the product of the undotted and dotted factors is 1; then $(\xi, \xi^\dagger) = -\sum |\xi_\alpha{}^i|^2 - \sum |\bar\xi_{\dot\alpha i}|^2 < 0$ and the dilatation part $\frac{1}{2}(\xi, \xi^\dagger)\Delta$ of $[\xi, \xi^\dagger]$ [prop:c2-special] is negative, as in (c). Labels: $(Q^i{}_\alpha)^\dagger = S^\alpha{}_i$, $(\bar Q_{\dot\alpha i})^\dagger = \bar S^{\dot\alpha i}$, $(P_{\alpha\dot\alpha})^\dagger = K^{\dot\alpha\alpha}$, $\Delta^\dagger = \Delta$, $(r^k{}_i)^\dagger = \overline{r^i{}_k}$ on $\mathfrak{sl}(W)$.

3 Two-point functions at order g^2

Target 8 (Form of the one-loop two-point function; $\mathbf{20'}$ and Konishi)**.** [S] [B04] (2.1)–(2.2), (2.14), (2.17)–(2.20):

$$\mu^{-2\delta D(g)} \langle O(x_1) O(x_2) \rangle = \frac{M_0}{|x_{12}|^{2D_0}} + g^2 \frac{M_2 + M_0 D_2 \log|\mu x_{12}|^{-2}}{|x_{12}|^{2D_0}} + \dots;$$

symmetric traceless Q_{mn}: $\langle Q_{mn} Q_{pq} \rangle = 2(1 - N^{-2}) \eta_{m(p}\eta_{q)n} (\Delta_{12}^2 + \frac{1}{4} g^2 \tilde H_{12,12}) + O(g^3)$, "the one-loop correction vanishes identically" at $\epsilon \to 0$, "intimately related to the vanishing of the beta function"; Konishi $ZK = (1 + \frac{1}{2} g^2 \gamma \xi_0/\epsilon) K$, $\langle ZK\, ZK \rangle = \frac{12(1-N^{-2})}{|x_{12}|^4}(1 + 6g^2 \log|\mu x_{12}|^{-2})$, $D = 2 + 6g^2 = 2 + \frac{3 g_{YM}^2 N}{4\pi^2}$, $g^2 = \frac{g_{YM}^2 N}{8\pi^2}$.

Regulator. DR/DRED (statement only); operators normal ordered.

Scheme. D_2 independent; M_2 dependent (choice of ξ_0 in Z); vanishing of the $\mathbf{20'}$ correction holds in this scheme, its scheme-independent form is Target 13.

Project object. $\langle a|b\rangle\big|_{\text{tree}}$ (M_0), $\langle a|b\rangle\big|_{\text{one loop}}$ (M_2), separation dependence $|x|^{-2(w_O + \text{coefficient})}$ fixed by D_1 (ch06 cor:c6-dilcomp); M_2 moves under the finite parts of $O(\varepsilon)$ (ch05 thm:c5-Oeps).

Two-component. $\mathbf{20'}$ and Konishi as in §1; $6g^2 = \frac{3 g_{YM}^2 N}{4\pi^2} = -24 \times \dfrac{\hbar\, \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X, Y)}$ (C7).

Target 9 (Pairwise structure of the order-g^2 correction)**.** [S] [DFS] (2.1)–(2.7): $\langle \mathrm{Tr}(z^1)^k(x)\,\mathrm{Tr}(\bar z^1)^k(y)\rangle = G(x,y)^k P_{k,k,0}(N)$, $G = 1/(4\pi^2(x-y)^2)$; two-particle interaction $(f^{pab}f^{pa'b'} + f^{pab'}f^{pa'b})B\,G^2$, $B = b_0 + b_1\ln(x-y)^2\mu^2$; self-energy $\delta^{aa'}NAG$, $A = a_0 + a_1\ln(x-y)^2\mu^2$; a_i, b_i gauge dependent; total order g^2:

$$\tfrac{1}{2}kN(B+2A)\sum_{\text{perms }\sigma}\mathrm{Tr}(T^{a_1}\cdots T^{a_k})\,\mathrm{Tr}(T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(k)}}), \qquad B+2A=0 \text{ by the } k=2 \text{ theorem},$$

using $\sum_p [[T^a,T^p],T^p] = NT^a$ ($\mathrm{tr}\,T^aT^b = \frac{1}{2}\delta^{ab}$; C7).

Regulator. unspecified; any regulator in which the $k=2$ statement holds.

Scheme. the ratio (order-g^2 part)/(tree) for $\mathrm{Tr}\,X^k$ equals $\frac{k}{2}$ times that for $k=2$ in every scheme with the pairwise structure; its vanishing needs Target 13.

Project object. [I] for $O = \int\omega\ \mathrm{tr}((\iota_w A)^k)$, $(w,w)=0$: $\langle O|O\rangle\big|_{\text{one loop}}/\langle O|O\rangle\big|_{\text{tree}} = \frac{k}{2}\times$(same for $k=2$) when the finite parts of $O(\varepsilon)$ are chosen pairwise (every one-loop graph with one O vertex meets exactly two legs of O, ch05 lem:c5-cycle); gauge dependence cancels in $B+2A$ (ch05, Feynman gauge only here).

Two-component. $z^1 \leftrightarrow \iota_w A = \frac{1}{4}w^{ij}A_{ij}$ with $w^{ij}w_{ij}=0$; $\mathrm{tr}(\iota_w A)^k = (-ig)^k(\iota_w A^a)\cdots\mathrm{tr}(T^{a_1}\cdots T^{a_k})$.

Target 10 (A position-space scheme: propagator smoothing)**.** [S] [Ryz] (13), (29)–(34) and the text after (34):

$$\tilde B(x,0) = -\tfrac{1}{4}Y^2\int\frac{(d^4z)\,(4\pi^2x^2)^2}{[4\pi^2((z-x)^2+\epsilon^2)]^2[4\pi^2(z^2+\epsilon^2)]^2} = -\frac{Y^2}{32\pi^2}\big(\log(x^2/\epsilon^2)-1\big), \quad Y^2 = 2g^2,$$

from $1/x^2 \to 1/(x^2+\epsilon^2)$ inside integrals ("simpler equivalent prescription" to differential regularisation; "the same result is obtained in dimensional regularization"); gauge-dependent $A(x,0) = 12\pi^2g^2\xi(\log x^2\mu^2 + \log 4\pi - \gamma) + (\xi\text{-independent})$ (DR), $B=-2A$, gauge-dependent part $\propto 2A+B=0$; $\langle O_i\bar O_j\rangle = [G]^{2p+q}(F_{ij}+\tilde B N G_{ij})$; pure operators $\langle Y_i\bar Y_j\rangle = C_i^0\delta_{ij}(1+\beta_i-\Delta_{1i}\log\mu^2x^2)/x^{2\Delta_0}$, "$\beta_j$ depends on the renormalization scale"; "diagonalize a symmetric matrix G of corrections with respect to the symmetric positive definite matrix F of free correlators", eigenvalues of $F^{-1}G$.

Regulator. position-space cutoff length ϵ on propagators (no DR in (13)); gauge dependence in (31) evaluated in DR.

Scheme. Δ_{1i} independent; β_i dependent.

Project object. statement only: the fixed ratio of the constant and the logarithm in (13), (33) is a property of the smoothing $1/x^2 \to 1/(x^2+\epsilon^2)$; (33) makes the constant vanish wherever the logarithm does. In the UV-boundary minimal scheme the **20′** one-loop constant is $\frac{27}{2} + 4\log\frac{3}{2}$ with vanishing logarithm (two_point_design thm:t-scalar), so the ratio does not hold there; after a field rescaling that removes the **20′** constant the trace part carries $\log|x|^2 + \gamma_E - 3\log 2$, against $\log(x^2/\epsilon^2) - 1$ in (13). No cross-check of the project's constants is drawn from this target.

Two-component. $z_i \leftrightarrow \iota_{w_i}A$, w_i null; $Y^2 = 2g^2 \leftrightarrow$ the quartic and Yukawa couplings of eq:c1-action written with $(X,Y) = 2\hbar X^aY^a$.

Target 11 (Konishi normalisation is free; point splitting)**.** [S] [BKRS] (17)–(19), (23)–(24), (27), (45) and text after (17): $K_1(x)|_{\text{reg}} = a(g^2)\sum_I :\mathrm{tr}(e^{-2gc(x)}\phi_I^\dagger(x+\frac{\epsilon}{2})e^{2gc(x)}\phi^I(x-\frac{\epsilon}{2})):$, $a = 1 + g^2a_1 + \ldots$; "Unlike the operators corresponding to symmetry generators (like the R-symmetry currents or the stress-energy tensor), the Konishi scalar K_1 has no intrinsic normalization"; $Z(\mu\epsilon) = (\mu^2\epsilon^2)^{-\gamma_{K_1}/2}$; $a_1 = 3N/8\pi^2$ makes the normalisation g-independent:

$$\langle K_1(x_1)K_1(x_2)\rangle = \frac{3(N^2-1)}{(4\pi^2)^2(x_{12}^2)^{2+\gamma_{K_1}(g^2)}}, \qquad \gamma_1 = \frac{3N}{4\pi^2};$$

(27) prints the log coefficient $-\frac{9N(N^2-1)}{4(4\pi^2)^3}$, $\frac{1}{4}$ of the expansion of (24) (C7; consistent with g^2 in (27) $= 4\times$ the standard g^2: footnote 3, the g of refs. [6], [7] is $2g$). (45): tree-level orthogonality $\langle O_iO_j\rangle_{\text{tree}} = \delta_{ij}N_i/(x_{ij}^2)^{\Delta_0}$, $N_i > 0$.

Regulator. point splitting of the composite operator, Fermi–Feynman gauge (position space, no DR).

Scheme. γ_1 independent; a_1 dependent ((18): a different splitting is a finite rescaling of a).

Project object. the finite part of the Konishi $O(\varepsilon)$ (ch05 thm:c5-Oeps, trace part) can set $\langle\text{Konishi}|\text{Konishi}\rangle\big|_{\text{one loop}} = 0$; not available for $\mathbf{20'}$ (Target 13).

Two-component. Konishi $= \frac{1}{4}\int\omega(A^{ij},A_{ij})$; $\gamma_1 g^2 = -24\times\dfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$.

Target 12 (Orthogonality of the Konishi and $\mathbf{20'}$ descendants)**.** [S] [BKRS] (9)–(11) and text after (11) (= `n4_untwisted` BKRS01 (9)–(13)); [EJSS] (13), (15), (17)–(21), footnote 13 (= `n4_untwisted` EJSS05; D6, D7; `q1_targets` T5):

$$(\mathcal{K}_{\mathbf{10}^*})_{AB} = 3\sqrt{2}g\,t^{ijk}_{AB}\,\mathrm{tr}(\varphi_i\varphi_j\varphi_k) + \tfrac{6g^2N}{32\pi^2}\,\mathrm{tr}(\bar\psi_{\dot\alpha A}\bar\psi^{\dot\alpha}_B),\quad (\mathcal{E}_{\mathbf{10}^*})_{AB} = -\,\mathrm{tr}(\bar\psi_{\dot\alpha A}\bar\psi^{\dot\alpha}_B) + \sqrt{2}g\,t^{ijk}_{AB}\,\mathrm{tr}(\varphi_i\varphi_j\varphi_k),$$

the anomalous term makes $\langle\delta^2K_1(x)\,\delta^2Q_{\mathbf{20'}}(y)\rangle = 0$; [EJSS]: $\langle\bar O_{\mathbf{20'}}O_{\mathbf{20'}}\rangle = 1/(x_{12}^2)^2 + \theta$ terms; $D^2O^{IJ}_{\mathbf{20'}/6} = -\frac{1}{2}(F_6^{IJ} - 4B_6^{IJ})$, unrenormalised in SSDR to g^2 (w.r.t. F, D^2O) and g^4 (w.r.t. B); $\langle\bar O_{\mathbf{10}/6}K_{\mathbf{10}/6}\rangle = 0$; footnote 13: whether the classical field equations receive corrections "may be a scheme-dependent property".

Regulator. [BKRS]: the anomalous coefficient of (9) is quoted from their ref. [8] (the Konishi anomaly), and the orthogonality is stated "as expected on the basis of the so-called $U_B(1)$ bonus symmetry [21]" (statements, no point-splitting computation of this entry); [EJSS] SSDR (statement only).

Scheme. orthogonality of eigen-operators of different dimension at separated points: independent; the bare form of the protected descendant: dependent ([EJSS] fn 13).

Project object. through one loop $\langle$(double ξ-image of Konishi)|(double ξ-image of $\mathbf{20'}$)$\rangle = 0$, both images built from the classical components and Q_1; the coefficient of the bilinear term is the Konishi entry T5 of `q1_targets`; in the inner product it pairs a length-two with a length-three letter word (Target 18).

Two-component. $\mathrm{tr}\,\bar\psi_{\dot\alpha A}\bar\psi^{\dot\alpha}{}_B \;\leftrightarrow\; (\bar\lambda_{\dot\alpha i}, \varepsilon^{\dot\alpha\dot\beta}\bar\lambda_{\dot\beta j})$ symmetrised in (ij) (**10** of $SU(4)$); $\mathrm{tr}\,\varphi\varphi\varphi \leftrightarrow (A_{ik},[A^{kl},A_{lj}])$ projected to the same **10** [I].

Target 13 (Normalisation of protected operators fixed by symmetry)**.** [S] [AFGJ] (2.6)–(2.7), text after (2.37), (4.15), introduction; [BRvR] crossing relation p. 2; [DFS] introduction; [EHW] (45)–(48) and conclusion; [HSW] abstract: $\langle J_\mu(x)J_\nu(0)\rangle = \frac{1}{16\pi^4}(\partial_\mu\partial_\nu - \delta_{\mu\nu}\partial^2)\frac{b(g(1/x))}{x^4}$; on the $\mathcal{N}=4$ fixed line "b is determined by the free field content"; primary central functions marginally constant; $v^2G(u,v)-u^2G(v,u)+4(u^2-v^2)+\frac{4(u-v)}{a} = 0$, $a = \dim G/4$ (protected part of the $\mathbf{20'}$ four-point function fixed by symmetry); "Certain 2- and 3-point correlators of operators in the same multiplet as the stress tensor and SU(4) flavor currents satisfy superconformal nonrenormalization theorems"; [EHW]: with $\hat A_q = g_{YM}^{-q}A_q$, $\partial_\tau\langle A^{q_1}\cdots A^{q_n}\rangle \sim \int d\mu\langle TA^{q_1}\cdots A^{q_n}\rangle$, and "the conjectured non-renormalisation theorem for 2 and 3 point correlation functions in $\mathcal{N}=4$ Yang-Mills theory holds exactly" (given analyticity and no nilpotent invariants for $n\le 4$); [HSW]: for the energy-momentum multiplet "anomaly considerations determine the coefficient".

Regulator. non-perturbative (anomaly matching [AFGJ]; harmonic-superspace Ward identities with assumptions [EHW], [HSW]; crossing [BRvR]).

Scheme. independent.

Project object. [I] the R-current and $T_{\mu\nu}$ are the Noether currents of $S^{(1)}(x_1)$ for x_1 in $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ and the conformal Killing fields; their two-point coefficients are the free ones; the $\mathbf{20'}$ is joined to them by ξ- and η-components, so $\langle\mathbf{20'}|\mathbf{20'}\rangle\big|_{\text{one loop}}$ is fixed by Q_1, S_1 (and the classical components) on the $\mathbf{20'}$ multiplet, not by a free finite part; the field-counting functional of ch06 implements [EHW]'s cancellation of the explicit powers of the coupling.

Two-component. R-current from $S^{(1)}$ with $r = \frac{1}{4}R_{IJ}\bar\Sigma^J\Sigma^I$ (dictionary `generators-even`); $\hat A_q$ with q letters $\iota_w A$, each carrying $-ig$ in $A_I = -igA_I^aT^a$.

Target 14 (Separated points versus contact terms)**.** [S] [PSZ] (2.3)–(2.5), abstract:

$$W[J,\bar J]\to\int d^4x_1d^4x_2d^4\theta\,J\,\frac{F(g^2,N)}{(x_1-x_2)^6}\,\bar J,\qquad \frac{1}{(x^2)^3}=\frac{\pi^{-2+\epsilon}}{64}\frac{\Gamma(-1-\epsilon)}{\Gamma(3)}\int d^np\,\frac{e^{-ipx}}{(p^2)^{-1-\epsilon}}$$

(momentum space, DR, MS): F is fixed by the $1/\epsilon$ poles, finite momentum-space terms are contact terms; order-g^4 corrections to $\langle\operatorname{Tr}\Phi^3\operatorname{Tr}\bar\Phi^3\rangle$ vanish for all N; at order g^2 by the stress-tensor-multiplet theorem.

Regulator. $\mathcal N=1$ superspace, DR + MS (statement only).

Scheme. separated-point coefficient independent; contact terms dependent.

Project object. $\langle a|b\rangle$ is a separated-point quantity; terms polynomial in the external momentum of the two-point function of $O_1[\infty],O_2[\infty]$ are contact terms and change with finite counterterms of the pair of insertions; only the non-polynomial part enters R5(b),(c) [I].

Two-component. $\operatorname{Tr}\Phi^3\leftrightarrow\int\omega\,\operatorname{tr}(\iota_wA)^3$, w null.

Target 15 (Quarter-BPS operators by tree-level orthogonality)**.** [S] [DHHR] abstract, introduction, (3.1): "the 1/4 BPS operators of the quantum theory are defined to be orthogonal to all the descendant operators with the same classical quantum numbers";

$$\tilde O\equiv O-\langle OL_i^\dagger\rangle_{\rm Born}\langle LL^\dagger\rangle^{-1}_{{\rm Born}\,ij}L_j,\qquad \langle\tilde OL_i^\dagger\rangle_{\rm Born}=0,\quad \langle\tilde OL_i^\dagger\rangle_{g^2}=\langle\tilde O'\tilde O^\dagger\rangle_{g^2}=0.$$

Regulator. that of [Ryz] (position space).

Scheme. protection independent; that no $O(g^2)$ admixture is needed: in this scheme.

Project object. the $\langle\,|\,\rangle\big|_{\rm tree}$-orthogonal complement of the image of the classical ξ-component (descendants) inside the classical BPS words equals the kernel of D_1 there (Target 27).

Two-component. L_i contain $\operatorname{Tr}[z_1,z_2]\cdots$, $z\leftrightarrow\iota_wA$.

Target 16 (Mixing of dimension-four $\mathbf{20'}$ operators; point splitting)**.** [S] [BERS] (26)–(29), (36)–(41) and text: scalars at the vertices of a tetrahedron of size ϵ; $O^{11}_\ell|_{\rm ren}=\lim\frac{a_\ell(g^2)}{(\epsilon^2)^{\gamma/2}}O^{11}_\ell|_{\rm reg}$, "$a_\ell\ldots$ depends on the regularization prescription"; $\langle O_i(x)O_j(y)\rangle=0$ for $i\neq j$ at tree level and at g^2; $\gamma_1^{O_i}=-\langle O_iO_i\rangle|_{g^2,\ln(x-y)^2}/\langle O_iO_i\rangle|_0$; finite corrections cancelled by the choice of $a_{\ell1}$; one real solution.

Regulator. point splitting (position space).

Scheme. γ_1 independent; a_ℓ dependent.

Project object. diagonal $\langle\,|\,\rangle\big|_{\rm one\ loop}$ removable by finite parts of $O(\varepsilon)$; off-diagonal orthogonality at tree level and one loop fixes the mixing; γ_1 is minus the log coefficient over the tree norm (ch06 cor:c6-dilcomp).

Two-component. four-scalar words in ι_wA projected to the $\mathbf{20'}$.

Target 17 (Classical non-orthogonality and completeness of the basis)**.** [S] [BKPSS] (3.6)–(3.8), (C.7): $\langle O^J_{ij,m}(x)\bar O^J_{ij,n}(0)\rangle=\frac{\delta_{mn}}{|x|^{2J+4}}(1+(\delta\Delta_n)L)$, $L=\log(x\Lambda)^{-2}$, "$\log\Lambda^2$ is a divergent constant (depending on the regularization scheme employed)" (DRED: $\frac{1}{\epsilon}+\gamma+\log\pi+2$); on the torus the classical part M^1_{mn} is not diagonal; (i) orthonormalising the classical part, (ii) diagonalising the quantum part by an orthogonal transformation is incorrect unless double-trace operators, whose overlap with single traces is $O(g_2)$, are included.

Regulator. DRED (statement only).

Scheme. $\delta\Delta_n$ independent; Λ dependent.

Project object. the operator basis for $\langle\,|\,\rangle\big|_{\rm tree}$-orthonormalisation must be complete in all trace structures ($\mathfrak g$ general in the project).

Two-component. $\operatorname{Tr}\phi_iZ^p\phi_jZ^{J-p}$ with $Z,\phi_i\leftrightarrow\iota_wA$ for mutually orthogonal w.

Target 18 (Order-g tree entries; the su(2|3) pair)**.** [D] (i) A tree graph with one cubic vertex joining operators of L and $L+1$ letters sends exactly one leg of the shorter and two legs of the longer operator

into the vertex; the remaining $L-1$ legs pair across. (ii) For $O_1 = \varepsilon^{\alpha\beta}\operatorname{Tr}\psi_\alpha\psi_\beta$, $O_2 = \varepsilon^{abc}\operatorname{Tr}\phi_a\phi_b\phi_c$ of [B03] (4.11)–(4.12) the remaining leg of $O_1^\dagger$ is a fermion and that of O_2 a scalar: $\langle O_1|O_2\rangle = 0$ at orders g^0 and g. (iii) With the [B03] §2.1 scalar product (cyclic classes orthogonal, square norm n for $|(A_1\dots A_{L/n})^n\rangle$): $\langle O_1|O_1\rangle = 4$, $\langle O_2|O_2\rangle = 18$; the matrix [B03] (4.13) with the entry $-9\sqrt{2}e^{-i\beta_2}\alpha_1^3 g^3$ (`n4_untwisted` D5) is self-adjoint for these norms (real α_1, β_2), the printed entry is not for $\beta_2 \notin \pi\mathbb{Z}$; the eigen-operators $O_1 + \frac{\sqrt{2}}{3}e^{i\beta_2}\alpha_1 g O_2$ and $O_2 - \frac{3}{\sqrt{2}}e^{-i\beta_2}\alpha_1 g O_1$ are orthogonal at order g (C2).

Proof. (i): k legs of the longer operator into the vertex, $L+1-k$ to the shorter, which then sends $L-(L+1-k) = k-1$ to the vertex; $k+k-1 = 3$. (ii): the leg of $O_1^\dagger$ is $\bar\psi$, the leg of O_2 is ϕ. (iii): $O_1 = 2\varepsilon^{12}|\psi_1\psi_2\rangle$, $O_2 = 3(|\phi_1\phi_2\phi_3\rangle - |\phi_1\phi_3\phi_2\rangle)$; self-adjointness $\langle O_i|HO_j\rangle = \langle HO_i|O_j\rangle$ evaluated symbolically.

Regulator. none (tree level; [B03] algebraic).

Scheme. independent within the [B03] frame family with real β_2.

Project object. in the su(2|3) sector the order-g length-changing terms sit in the classical ξ-, η-components (the cubic term of $S^{(1)}(\xi)$, prop:c7-susy) and in D_1, not in $\langle\,|\,\rangle\big|_{\text{tree}}$; independent confirmation of the erratum D5.

Two-component. su(2|3) letters ϕ_a, $\psi_\alpha \leftrightarrow$ scalar letters A_{a4} and undotted fermion letters for $w = e_4$, as in `q1_targets` (dictionary `letters`) [I].

4 Hermiticity of the one-loop dilatation operator; the inner product beyond tree level

Target 19 (Self-adjoint one-loop vertex; symmetric renormalisation)**.** [S] [BKS] (2.3)–(2.13), (8.11), (C.17)–(C.18) and comments after (C.18): $V_2(x) =: W_2(x, I_{0x}^{-1}\Phi, \check\Phi)$:, "V_2 is self-adjoint with respect to the tree-level scalar product"; $\tilde O = (1 - \frac{1}{2}g^2 V_2(x_0))O$; $V_2(x) = \frac{\Gamma(1-\epsilon)}{(\frac{1}{2}\mu^2x^2)^{-\epsilon}}V_2$, $D_2 = -\lim \epsilon V_2$,

$$\langle \tilde O_\alpha^+ \tilde O_\beta^- \rangle_{\text{one-loop}} = \exp(W_0)\exp\big(\log(x_0^2/x^2)\, g^2 D_2^-\big) O_\alpha^+ O_\beta^-\big|_{\Phi=0},$$

"the coefficient of the correlator is given by its tree-level value as D_2 multiplies the logarithm only"; four loops: $D' = (1+\beta' g^6[D_2, D_4])^{-1} D (1+\beta' g^6[D_2, D_4])$, orthogonal (antisymmetric generator), eigenvalues unchanged; "V_{2l} are symmetric with respect to the scalar product induced by free contractions … The same holds for the dilatation generator which consequently has real eigenvalues".

Regulator. DR (statement only).

Scheme. eigenvalues independent; vanishing of the one-loop coefficient is the scheme choice $\tilde O$.

Project object. a scheme with $\langle a|b\rangle\big|_{\text{one loop}} = 0$ exists formally; in it the order-$\hbar$ identity of Proposition 1 reads $\langle x_1\text{-component}_1(a)|b\rangle\big|_{\text{tree}} = \langle a|x_1^\dagger\text{-component}_1(b)\rangle\big|_{\text{tree}}$ for every generator: D_1, Q_1 and S_1, P_1 and K_1 are adjoint pairs for $\langle\,|\,\rangle\big|_{\text{tree}}$.

Two-component. $g^2_{[\text{BKS}]} = g^2_{YM}/16\pi^2$; D_2 on $\mathbf{20'}$ and Konishi: 0 and $-24 \times \dfrac{\hbar\,\operatorname{tr}_{\mathfrak g}(\operatorname{ad}_X \operatorname{ad}_Y)}{16\pi^2(X,Y)}$ in the normalisation of ch06.

Target 20 (All orders: the classical scalar product can be kept)**.** [S] [B04] (2.25), (2.39)–(2.43), (2.117), (2.121)–(2.136), §2.6.5 (2.156)–(2.158): $D_{\text{ren}} = Z^{-1} D_{\text{bare}} Z$; pre-diagonalisation $J(g) \mapsto T(g)J(g)T^{-1}(g)$, $T = 1 + \sum_{d\neq 0}\frac{1}{d}\delta D_d + \dots$; $V_{2\ell}^T = \dot V_{2\ell}$ "shows that $V_{2\ell}$ is self-adjoint"; $Z = \exp\big(-\frac{1}{2}\sum g^{2\ell}V_{2\ell}(1/\mu) + \frac{1}{24}g^6[V_2(1/\mu), V_4(1/\mu)] + \dots\big)$; $\langle ZO\,\dot Z\dot O\rangle = \exp(W_0)\exp(\log|\mu x_{12}|^{-2}\sum g^{2\ell}D_{2\ell})O\dot O$, $D_{2\ell} = -\ell\lim\epsilon V_{2\ell}$; "the programme ensures that the coefficient of the two-point function is given by free contractions of the unrenormalised operators"; §2.6.5: "states can be renormalised in such a way as to preserve the classical scalar product", planar scalar product (2.156), $\|\,|(A_1\dots A_{L/n})^n\rangle\| \sim \sqrt{n}$, adjoint $\binom{A_1\dots A_{E_i}}{B_1\dots B_{E_o}}^\dagger \sim \binom{B_{E_o}\dots B_1}{A_{E_i}\dots A_1}$; a self-adjoint interaction is a Hermitian matrix only on normalised states, e.g. the asymmetric (2.117).

Regulator. DR (statement only).

Scheme. existence of the frame is a statement about schemes; spectrum independent.

Project object. the frame of Target 19 to all orders; the commutator term in Z is the four-loop analogue.

Two-component. as in Target 19.

Target 21 (su(2|3): hermiticity of $Q_2^{[\mathrm{B03}]}$, $S_2^{[\mathrm{B03}]}$ and the similarity parameters)**.** [S] [B03] §2.1 "Adjoint", (3.9), (3.10)–(3.13), (3.36)–(3.40) and the following paragraph, Table 1, (4.4)–(4.7): "We work with a complex algebra and will not require an adjoint operation here"; $\binom{A_1\dots A_n}{B_1\dots B_m}^\dagger \sim \binom{B_1\dots B_m}{A_1\dots A_n}$; $(Q_0)^a{}_\alpha = e^{i\beta_1}\binom{a}{\alpha}$, $(S_0)^\alpha{}_a = e^{-i\beta_1}\binom{\alpha}{a}$; $c_9 = \frac{1}{\sqrt{2}}\alpha_1 e^{i\beta_1+i\beta_2}$, $c_{10} = \frac{1}{\sqrt{2}}\alpha_1 e^{-i\beta_1-i\beta_2}$; "In a real form ... the algebra should be self-adjoint which imposes some reality constraint on $\alpha_1, \beta_{1,2}$. For real su(2|3) they would have to be real and α_1^2 would be positive"; $J(g) \mapsto T(g)J(g)T(g)^{-1}$, $T = 1 + g^2 T_2$, parameters $\gamma_{1..4}$ in $Q_2^{[\mathrm{B03}]}$, $S_2^{[\mathrm{B03}]}$ (Table 1, transcribed in `n4_untwisted` B03).

[D] (C1) Let the adjoint map the k-th structure of $Q_2^{[\mathrm{B03}]}$ to s_k times the k-th structure of $S_2^{[\mathrm{B03}]}$ (order of Table 1, $s_k = \pm 1$). Since the γ-independent sums of coefficients are $\frac{\alpha_1^2}{2}(1, 0, -1, 1)$, $S_2^{[\mathrm{B03}]} = (Q_2^{[\mathrm{B03}]})^\dagger$ forces $s_1 = s_3 = s_4 = +1$ and $\gamma_1, \gamma_2, \gamma_3 - \gamma_4 \in \mathbb{R}$; the row swap (3.9) ($s_2 = +1$) gives all $\gamma_k \in \mathbb{R}$. The frames $Q_2^{[\mathrm{B03}]} = 0$, $\gamma = \frac{i\alpha_1^2}{4}(1, 1, -1, 1)$, and $S_2^{[\mathrm{B03}]} = 0$ (`n4_untwisted` D2) are not hermitian frames for the tree-level adjoint.

Regulator. none (algebraic; [B04] §5.3.3: classical su(3)×su(2) kept).

Scheme. the γ's are similarity (scheme) parameters; hermiticity selects real γ given the tree-level adjoint; the non-hermitian frames need a corrected scalar product (Proposition 1).

Project object. the one-loop ξ- and η-components Q_1, S_1 (user's names; the [B03] counting Q_2 is the project's one-loop order, `sweep.json` cross-paper finding) can be moved between ξ and η by finite parts of $O(\varepsilon)$ only at the price of a nonzero $\langle\,|\,\rangle\big|_{\text{one loop}}$.

Two-component. $a \leftrightarrow$ scalar letters A_{a4}, $\alpha \leftrightarrow$ fermion letters (`q1_targets letters`) [I]; [B03] ε signs absorbed by $\beta_2 \to \beta_2 + \pi$ (`n4_untwisted` D3).

Proposition 1 (One-loop hermiticity identity)**.** [D] Let the generators act on a space of local operators with matrices $X(\hbar) = X|_{\hbar^0} + \hbar X|_{\hbar^1}$ in a basis, let the Gram matrix of $\langle\,|\,\rangle$ be $\langle\,|\,\rangle\big|_{\text{tree}}$ times $(1 + \hbar\,(\text{tree Gram})^{-1}(\text{one-loop Gram}))$, and let $Y^{\text{adj}} = (\text{tree Gram})^{-1}\,\overline{Y}^{\mathrm{T}}$ (tree Gram) be the tree-level adjoint. Hermiticity $\langle Xa|b\rangle = \langle a|X^\dagger b\rangle$ through order $\hbar$ is equivalent to

$$X^\dagger|_{\hbar^0} = (X|_{\hbar^0})^{\text{adj}}, \qquad X^\dagger|_{\hbar^1} = (X|_{\hbar^1})^{\text{adj}} + \big[X^\dagger|_{\hbar^0},\ (\text{tree Gram})^{-1}(\text{one-loop Gram})\big].$$

Under the change of every operator $a \mapsto a + \hbar\,(\text{a linear map applied to } a)$ the one-loop generator matrix changes by the commutator of that map with the tree one, $(\text{tree Gram})^{-1}(\text{one-loop Gram})$ changes by minus the map minus its tree adjoint, and the identity is preserved. In particular, with the η-component along $\xi^\dagger$ as $Q^\dagger$ and the special conformal component as $P^\dagger$:

$$\begin{aligned}
S_1 &= Q_1^{\text{adj}} + \big[\text{classical } \eta\text{-component},\ (\text{tree Gram})^{-1}(\text{one-loop Gram})\big],\\
K_1 &= P_1^{\text{adj}} + \big[\text{classical special conformal component},\ (\text{tree Gram})^{-1}(\text{one-loop Gram})\big],\\
D_1 &= D_1^{\text{adj}} + \big[\text{classical dilatation},\ (\text{tree Gram})^{-1}(\text{one-loop Gram})\big],
\end{aligned}$$

so D_1 is self-adjoint for $\langle\,|\,\rangle\big|_{\text{tree}}$ exactly when the one-loop Gram matrix commutes with the classical dilatation (e.g. in the frame of [B04] (2.39)).

Proof. $X^\dagger = (\text{Gram})^{-1}\,\overline{X}^{\mathrm{T}}\,(\text{Gram})$ with $\text{Gram} = (\text{tree Gram})\big(1 + \hbar\,(\text{tree Gram})^{-1}(\text{one-loop Gram})\big)$; the first-order expansion of

$$\begin{aligned}
&\big(1 + \hbar(\text{tree Gram})^{-1}(\text{one-loop Gram})\big)^{-1}(\text{tree Gram})^{-1}\big(\overline{X|_{\hbar^0}}^{\mathrm{T}} + \hbar\overline{X|_{\hbar^1}}^{\mathrm{T}}\big)\\
&\quad\times(\text{tree Gram})\big(1 + \hbar(\text{tree Gram})^{-1}(\text{one-loop Gram})\big)
\end{aligned}$$

is the displayed pair. Under $a \mapsto (1+\hbar\,\mathrm{map})\,a$: $X \mapsto (1+\hbar\,\mathrm{map})X(1+\hbar\,\mathrm{map})^{-1}$, Gram $\mapsto$ $\overline{(1+\hbar\,\mathrm{map})^{-1}}^{\mathrm{T}}\,\mathrm{Gram}\,(1+\hbar\,\mathrm{map})^{-1}$, so (tree Gram)$^{-1}$(one-loop Gram) changes by $-\mathrm{map}-\mathrm{map}^{\mathrm{adj}}$ and $X|_{\hbar^1}$ by $[\mathrm{map}, X|_{\hbar^0}]$; substituting, both sides of the second equation change by $[\mathrm{map}, X^\dagger|_{\hbar^0}]$. Checked on random matrices (C3). □

Project object (R5(c)). for $x_1 = \xi$ and every pair a, b:

$$\langle Q_1(a)|b\rangle\big|_{\text{tree}} - \langle a|S_1(b)\rangle\big|_{\text{tree}} = \langle a|\{S^{(1)}(\xi^\dagger), b\}\rangle\big|_{\text{one loop}} - \langle\{S^{(1)}(\xi), a\}|b\rangle\big|_{\text{one loop}},$$

with $\xi^\dagger$ of Target 7, $\langle\,|\,\rangle\big|_{\text{one loop}}$ from two-point graphs, Q_1, S_1 from the equivariant cochain (evaluation signs of def:o-comp; for odd operators the Hilbert-space adjoint carries no extra sign, the transcription to cochain components is part of R5(c)); likewise for (P_1, K_1), (D_1, D_1), (M_1, M_1), (R_1, R_1). Finite parts of $O(\varepsilon)$ act as the linear map above; arity-zero, arity-one and arity-two finite counterterms and the choice of Q^{GF} act through the same covariance when they change $O[\infty]$ by local terms [I].

Two-component. $\xi = (\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$, $\xi^\dagger = (-\overline{\xi_\alpha{}^i}, \overline{\bar\xi_{\dot\alpha i}}) \in \Sigma^*$.

Target 22 (Frame with uncorrected translations)**.** [D] (C4) In a frame with $P_1 = 0$ (ch07: translations covariant), Proposition 1 with $P^\dagger = K$ gives, with the order-$\hbar$ part of $[K, P] = 2D + (\text{rotation})$ ([SD] (116) normalisation), on states of a lowest-weight module

$$K_1 = \big[K|_{\hbar^0}, (\text{tree Gram})^{-1}(\text{one-loop Gram})\big],$$
$$2D_1 = \Big[\big[K|_{\hbar^0}, (\text{tree Gram})^{-1}(\text{one-loop Gram})\big], P|_{\hbar^0}\Big].$$

Conformal sl(2) with lowest weight $w_O + \hbar \times (\text{coefficient})$, $P|m\rangle = |m+1\rangle$: Gram $m!\,(2w_O + 2\hbar \times \text{coefficient})_m$, ratio of one-loop to tree part on level m: $\sum_{j=0}^{m-1} \frac{2\hbar\times(\text{coefficient})}{2w_O+j}$; level one minus level zero: $\hbar \times (\text{coefficient})/w_O$, from $\langle O|K_\mu P_\nu|O\rangle = 2\Delta^{[\mathrm{SD}]}\delta_{\mu\nu}\langle O|O\rangle$ [SD] (116) in four dimensions ($\Delta^{[\mathrm{SD}]}$ the scaling dimension).

[S] [Z05] (3.6)–(3.7), (2.19) (`n4_untwisted` Z05, D4): $\mathfrak{X}_2^\pm = \pm[\mathfrak{X}_0^\pm, \mathfrak{x}] + [\mathfrak{X}_0^\pm, \mathfrak{y}]$; for $\mathfrak{y} = -\mathfrak{x}$ the raising generators are uncorrected and $\mathfrak{K}_2^{22} \neq 0$; hermiticity $\mathfrak{X}^+ = (\mathfrak{X}^-)^\dagger$ is lost for $\mathfrak{y} = -\mathfrak{x}$; $\overleftarrow{\mathfrak{T}}_1^-$ from $\overrightarrow{\mathfrak{T}}_1^+$ "by hermitian conjugation".

Regulator. none (algebraic).

Scheme. the level-one minus level-zero ratio is independent within frames with $P_1 = 0$.

Project object. with $P_1 = 0$, K_1 is fixed by the one-loop Gram matrix on P-descendants; cross-check of odd_components_design thm:o-KD (under audit) and of the special conformal component of ch07 rem:c7-KDelta.

Two-component. $[P_{\alpha\dot\alpha}, K^{\dot\beta\beta}]$ via $[x \mapsto v, (x,x)b - 2(b,x)x] = -(b,v)\Delta +$ rotation (Target 6).

Target 23 (Hermitian conjugation of oscillators; non-unitary length similarity)**.** [S] [Zw11] introduction, (2.5), end of §2.1, (2.12)–(2.15): inner product $\langle \vec n_1 \dots \vec n_{L'}|\vec m_1 \dots \vec m_L\rangle = \delta_{LL'}\prod_i \delta(\vec n_i - \vec m_i)$ without cyclic sum; radially "Hermitian conjugation simply interchanges creation and annihilation oscillators. For example, $Q^\dagger = S$, and $D^\dagger = D$"; "Because the dilatation generator is Hermitian, $D_{2\to L}$ follow from Hermitian conjugation of $D_{L\to 2}$"; $J_{L\to L'}(g) = \sum_l g^{L+L'+2l-2} J^{(l)}_{L\to L'}$; similarity $J \to g^L J g^{-L}$ (same eigenvalues) gives $J_{L\to L'}(g) = g^{2L'-2}J_{L\to L'}$; scheme $[D_0, J(g)] = \dim(J)J(g)$; $L, \dot L, R$ uncorrected.

Regulator. none (algebraic; tree amplitudes).

Scheme. eigenvalues independent; hermiticity frame dependent.

Project object. the coupling sits in $(-,-)$ (rem:c1-real), a rescaling of every letter by a power of the coupling relative to canonical fields: $\langle\,|\,\rangle\big|_{\text{tree}}$ and the generator matrices of Proposition 1 must be taken in the same frame.

Two-component. $(X, Y) = 2\hbar X^a Y^a$, $A_I = -igA_I^a T^a$: each letter carries one power of g.

Target 24 (Wick matrix and self-adjoint one-loop Hamiltonian)**.** [S] [BMV] (3), (6)–(7), text before and after (9): $H = \sum C_{AC}^{BD}(-)^{BC}(-)^{BD} : \mathrm{tr}[W_A, \check W^B\}[W_C, \check W^D\} :$; $W_{ij} = \langle v_i(0)^\dagger v_j(1)\rangle$; "the dilatation

operator is self-adjoint with respect to the inner product (6)", $\langle v_i^\dagger H v_j\rangle = \langle (Hv_i)^\dagger v_j\rangle$, i.e. $WH = (WH)^\dagger$; $H = \{Q, Q^\dagger\}$ positive; at non-integer N non-hermitian H, complex dimensions, negative norms; null space of W = trace relations.

Regulator. none (one-loop Hamiltonian taken from [B03c]).

Scheme. independent (spectrum).

Project object. $\langle\,|\,\rangle\big|_{\text{tree}}$ for general $\mathfrak{g}$ includes trace relations (degenerate Gram); D_1 self-adjoint for it (Proposition 1 with commuting one-loop Gram).

Two-component. letters W_A of the field-strength module $\leftrightarrow$ jets of $A_m, A_{ij}, \lambda_\alpha{}^i, \bar\lambda_{\dot\alpha i}$.

5 The 1/16-BPS and holomorphic-twist literature

Target 25 ($S = Q^\dagger$; Hodge correspondence; one-loop Hamiltonian)**.** [S] [GGKM] (2.1), §2, introduction (= `n4_untwisted` GGKM08); [CY] (2.1), (5.3)–(5.6) (= CY13); [CL22] (2.1), footnote 2, (3.1)–(3.3), footnotes 8–10, §4.3 (= CL22; `q1_targets` T7):

$$S_{i\alpha} = (Q^i{}_\alpha)^\dagger, \quad \bar S^{i\dot\alpha} = (\bar Q_{i\dot\alpha})^\dagger, \quad \bar Q_{i\dot\alpha} = (Q^i{}_\alpha)^*, \quad \bar S^{i\dot\alpha} = (S_{i\alpha})^*;$$
$$2\{Q, Q^\dagger\} = \Delta^{[\mathrm{CY}]} \equiv E - 2J - H_1 - H_2 - H_3;$$

"states that satisfy $\{Q_{4-}, S_{4-}\}|\Psi\rangle = 0$ are in one-to-one correspondence with states in the cohomology of Q_{4-}"; conjecture: the supersymmetric spectrum equals the 1/16-BPS states of the finite-N one-loop Beisert Hamiltonian; [CY]: $Q^{\mathrm{ad}} = \mathrm{Tr}\left(\Psi \frac{\partial}{\partial\Psi}\frac{\partial}{\partial\Psi}\right)$ for the Fock inner product, $H^*_Q(V_N) \cong \ker\{Q, Q^{\mathrm{ad}}\}$, Q^{ad} does not commute with π; [CL22]: "A 1/16-BPS operator must also be annihilated by $Q^\dagger$, which receives quantum corrections"; "Q-exact operators are orthogonal to the 1/16-BPS operators"; $D = D_0 + \frac{g_{YM}^2 N}{8\pi^2} H$, $H \propto \{Q, Q^\dagger\}$ on BPS words, $Q = \mathrm{Tr}\,\Psi\Psi\frac{\partial}{\partial\Psi}$, $Q^\dagger = \mathrm{Tr}\,\Psi\frac{\partial}{\partial\Psi}\frac{\partial}{\partial\Psi}$ at leading order; "Demanding the annihilation by the one-loop $Q^\dagger$ (3.2) will give the weak coupling of the 1/16-BPS operator".

Regulator. none (algebraic; free inner product).

Scheme. cohomology independent; the BPS representative of a class is scheme and frame dependent.

Project object. the classical ξ-component on BPS words for $\xi = u \otimes w$ of rank one (`q1_targets` dictionary) and its $\langle\,|\,\rangle\big|_{\text{tree}}$-adjoint, the η-component along $\xi^\dagger$ (Target 7); the one-loop representative from Proposition 1 with $x_1 = \xi^\dagger$.

Two-component. $Q = Q^4{}_- \leftrightarrow \xi_\alpha{}^i = u_\alpha w^i$, $\bar\xi = 0$; $S = Q^\dagger \leftrightarrow \eta^\alpha{}_i = -\overline{u_\alpha}\,\overline{w^i}$, $\bar\eta = 0$.

Target 26 (Unique S-closed representative; su(1|1) and su(2|1); coupling counting)**.** [S] [BK] footnote 4: "1/4N-BPS operators are by definition annihilated by one Q and $S = Q^\dagger$. In each Q-cohomology class there is a unique representative that is also annihilated by S"; [BGKWWY] (2.4)–(2.5): $\{Q_{\dot\pm}, S_{\dot\pm}\} = \Delta + \frac{3}{2}R - M_{\dot+\dot-}$, $\{Q_{\dot\alpha}, S_{\dot\beta}\} = (\Delta + \frac{3}{2}R)\epsilon_{\dot\alpha\dot\beta} - M_{\dot\alpha\dot\beta}$; [KKLP] (2.13) and text: $Q = Q_{\text{free}} + g_{YM}Q_0 + g_{YM}^2 Q_1 + O(g_{YM}^3)$, "$H = \{Q, Q^\dagger\} \sim g_{YM}^2$ is one-loop", field redefinition $\Phi \to g_{YM}^{-1}\Phi$; [KLLS] p. 5: "the classical supercharge Q is at half-loop level since $H = \{Q, Q^\dagger\} \sim g_{YM}^2$".

Regulator. [BK], [BGKWWY]: Schwinger-time cutoff (twisted, `sweep.json`); [KKLP], [KLLS]: algebraic.

Scheme. independent statements.

Project object. [KKLP]'s $\Phi \to g_{YM}^{-1}\Phi$ is the frame change of Target 23 (coupling in $(-,-)$); twisted Q_1 of [BK] = one-loop order of the project (`q1_targets` T1).

Two-component. $Q_{\dot-}$ of [BGKWWY] is an $\mathcal{N} = 1$ label; for $\mathcal{N} = 4$ see Target 25.

Target 27 (One-loop dilatation on BPS words from the classical supercharge)**.** [D] Let $2\{Q, Q^\dagger\}$ be the exact relation of Target 25 (equivalently [KMMR] (3.1), [B03] (4.3)), expand $Q = Q|_{g^0} + gQ|_{g^1} + g^2 Q|_{g^2} + \dots$ in the [B03] counting, and let Π project onto classical BPS words ($Q|_{g^0}$ and $Q^\dagger|_{g^0}$ vanish there). Then in every frame

$$\Pi\,\delta D|_{g^2}\,\Pi = 2\big\{\Pi Q|_{g^1}\Pi,\ (\Pi Q|_{g^1}\Pi)^{\mathrm{adj}}\big\},$$

with adj the adjoint for $\langle\,|\,\rangle\big|_{\text{tree}}$: the one-loop dilatation on BPS words is fixed by the classical length-changing supercharge and the tree inner product; its kernel is $\ker \Pi Q|_{g^1}\Pi \cap \ker(\Pi Q|_{g^1}\Pi)^{\text{adj}}$, the tree-orthogonal complement of the image (Target 15).

Proof. Only D is corrected in $2\{Q, Q^\dagger\} = \Delta^{[\text{CY}]} = D - 2J - \dots$, so $\delta D|_{g^2} = 2(\{Q|_0, Q^\dagger|_2\} + \{Q|_1, Q^\dagger|_1\} + \{Q|_2, Q^\dagger|_0\})$. $Q|_0\Pi = Q^\dagger|_0\Pi = 0$ (free supercharges annihilate BPS letters), and $\Pi Q|_0 = \Pi Q^\dagger|_0 = 0$ because $Q^\dagger|_0 = (Q|_0)^{\text{adj}}$ and Π is the tree-orthogonal projector onto an eigenspace of $\Delta^{[\text{CY}]}|_0$; so the first and last terms drop. $[\Delta^{[\text{CY}]}|_0, Q|_1] = [\Delta^{[\text{CY}]}|_0, Q^\dagger|_1] = 0$ (order g of $[\Delta^{[\text{CY}]}, Q] = 0$, $\delta D|_g = 0$), so $Q|_1$, $Q^\dagger|_1$ preserve $\operatorname{im}\Pi$ and its tree-orthogonal complement, and $\Pi\{Q|_1, Q^\dagger|_1\}\Pi = \{\Pi Q|_1\Pi, \Pi Q^\dagger|_1\Pi\}$. By Proposition 1 at order g, $Q^\dagger|_1 = (Q|_1)^{\text{adj}} + \big[Q^\dagger|_0, (\text{tree Gram})^{-1}(\text{order-}g\text{ Gram})\big]$, and $\Pi\big[Q^\dagger|_0, \text{any matrix}\big]\Pi = 0$.

Regulator. none.

Scheme. independent at this order.

Project object. in canonical normalisation $Q|_{g^1}$ is the classical nonlinear part of the ξ-component, from the cubic term $\frac{1}{2}(\rho(\frac{1}{2}[A \wedge A])\xi, \lambda^*)$ of $S^{(1)}(\xi)$ (prop:c7-susy); $\delta D|_{g^2}$ on BPS words is the dimension shift whose double is D_1 there (ch06): a test of D_1 by classical data and $\langle\,|\,\rangle\big|_{\text{tree}}$ only [I].

Two-component. [CL22] (3.2) $Q = \operatorname{Tr}\Psi\Psi\partial_\Psi$ with Ψ the BPS superfield of [CY] (2.9) $\leftrightarrow$ $\xi = u \otimes w$ letters (`q1_targets` dictionary `letters`) [I].